ECE 329 Exam 2 · FRONT $\mu_0 = 4\pi\times10^{-7}$ | $\epsilon_0 \approx 8.85\times10^{-12} \approx \frac{10^{-9}}{36\pi}$ | $c = 3\times10^{8}$ | $\eta_0 = 120\pi$

Maxwell

$\oint \mathbf{D}\cdot d\mathbf{S} = Q_{enc}$ | $\nabla\cdot\mathbf{D} = \rho$

$\oint \mathbf{B}\cdot d\mathbf{S} = 0$ | $\nabla\cdot\mathbf{B} = 0$

$\oint \mathbf{E}\cdot d\mathbf{l} = -\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{S}$ | $\nabla\times\mathbf{E} = -\partial_t\mathbf{B}$

$\oint \mathbf{H}\cdot d\mathbf{l} = I_{enc} + \frac{d}{dt}\int \mathbf{D}\cdot d\mathbf{S}$ | $\nabla\times\mathbf{H} = \mathbf{J} + \partial_t\mathbf{D}$

$\nabla\cdot\mathbf{J} = -\partial_t\rho$ | $\mathbf{J} = \sigma\mathbf{E}$ | $\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$

Materials

$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P} = \epsilon\mathbf{E}$, $\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$, $\epsilon_r = 1+\chi_e$

$\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M}) = \mu\mathbf{H}$, $\mathbf{M} = \chi_m\mathbf{H}$, $\mu_r = 1+\chi_m$

dia $\chi_m < 0$ (tiny), para > 0 (tiny), ferro $\gg 1$

bound: $\rho_b = -\nabla\cdot\mathbf{P}$, $\mathbf{P}\cdot\hat{n}$; $\mathbf{J}_b = \nabla\times\mathbf{M}$, $\mathbf{M}\times\hat{n}$

Boundaries ($\hat{n}$: side − → side +)

E_t, B_n continuous | $\hat{n}\cdot(\mathbf{D}^+ - \mathbf{D}^-) = \rho_s$

$\mathbf{H}^+ - \mathbf{H}^- = \mathbf{J}_s\times\hat{n}$ (free J_s only)

no ρ_s/J_s: D_n, H_t continuous, so E_n, B_t scale by ϵ, μ ratio

conductor: $\mathbf{E} = 0$, $\rho = 0$ inside; $\rho_s = \hat{n}\cdot\mathbf{D}$ outside

H of shapes (Ampère, RHR)

wire $\frac{I}{2\pi r}\hat{\phi}$ | thick wire inside $\frac{Ir}{2\pi a^2}$

coax $\frac{I}{2\pi r}$ $(a<r<b)$, 0 outside

solenoid $H = \frac{N}{\ell}I$ inside, 0 outside | toroid $\frac{NI}{2\pi r}$

sheet $\mathbf{H} = \frac{1}{2}\mathbf{J}_s\times\hat{n}$ ($\hat{n}$ sheet → point); ± sheet pair: J_s between, 0 out

slab: 0 at mid-plane, linear inside, $\pm\frac{JW}{2}$ outside

loop axis $B = \frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}}$, center $\frac{\mu_0 I}{2a}$

$d\mathbf{B} = \frac{\mu_0 I\, d\mathbf{l}\times\hat{R}}{4\pi R^2}$ | $d\mathbf{F} = I\, d\mathbf{l}\times\mathbf{B}$

wires $\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d}$: parallel attract

$I\delta(x)\delta(y)\hat{z}$ = wire (A) | $J_s\delta(z)$ = sheet (A/m)

check a shape with the curl: $H_\phi(r)\hat{\phi} \to J_z = \frac{1}{r}\frac{d(rH_\phi)}{dr}$ | $H_z(r)\hat{z} \to J_\phi = -\frac{dH_z}{dr}$

Faraday

$\mathcal{E} = \oint(\mathbf{E}+\mathbf{v}\times\mathbf{B})\cdot d\mathbf{l} = -N\frac{d\Psi}{dt}$, $\Psi = \int\mathbf{B}\cdot d\mathbf{S}$

fingers along C → thumb = $d\mathbf{S}$. Lenz is a check, not a 2nd minus.

bar $vB\ell$ | rails $I = \frac{vB\ell}{R}$, $F = I\ell B$ against $\mathbf{v}$

spinning loop $\Psi = BA\cos\omega t \to \mathcal{E} = BA\omega\sin\omega t$

uniform $B_z(t)$, radius a: $E_\phi = -\frac{r}{2}\dot{B}$ in, $-\frac{a^2}{2r}\dot{B}$ out

meter reads $\int_+^- \mathbf{E}\cdot d\mathbf{l}$ along its leads; around a loop $\sum IR = \mathcal{E}_{\text{enclosed}}$; leads hugging R: $\pm IR$

Inductance

$L = \frac{N\Psi}{I}$ | $v = L\frac{dI}{dt}$ | $W = \frac{1}{2}LI^2 = \int\frac{1}{2}\mu H^2 dV$

solenoid $\frac{\mu N^2 A}{\ell}$ | toroid $\frac{\mu N^2 h}{2\pi}\ln\frac{b}{a}$

coax $L' = \frac{\mu}{2\pi}\ln\frac{b}{a}$ | pp line $L' = \frac{\mu d}{w}$ | $L'C' = \mu\epsilon$

mutual $M = \frac{N_2\Psi_{21}}{I_1}$

N turns: N/ℓ in H and N again in the linkage: $\mathcal{E}, L \propto N^2$

ECE 329 Exam 2 · BACK $v_p = \frac{1}{\sqrt{\mu\epsilon}}$ | $\eta = \sqrt{\mu/\epsilon}$ | $c = 0.3$ m/ns

Statics (Exam 1)

E: point $\frac{Q}{4\pi\epsilon_0 r^2}$, line $\frac{\rho_l}{2\pi\epsilon_0 r}$, sheet $\frac{\rho_s}{2\epsilon_0}$, slab $\frac{\rho_v}{\epsilon_0}$ in

$V_A - V_B = \int_A^B \mathbf{E}\cdot d\mathbf{l}$ | $\mathbf{E} = -\nabla V$ | $\nabla^2 V = -\rho/\epsilon$

C, G, relaxation

$C = Q/V$: pp $\frac{\epsilon A}{d}$ | coax $\frac{2\pi\epsilon\ell}{\ln(b/a)}$ | spheres $\frac{4\pi\epsilon ab}{b-a}$ | sphere $4\pi\epsilon a$

stacked: D same, $V = D\sum \frac{d_i}{\epsilon_i}$ | side by side: E same, C's add

$R = \frac{\ell}{\sigma A}$ | $\frac{G}{C} = \frac{\sigma}{\epsilon}$ | $RC = \frac{\epsilon}{\sigma}$

$\rho = \rho_0 e^{-t/\tau}$, $\tau = \epsilon/\sigma$; E from ρ: $\partial_z E_z = \rho/\epsilon$

Displacement current

$\mathbf{J}_d = \partial_t \mathbf{D}$ | $I_d = \frac{d\Psi_e}{dt}$ | $\Psi_e = \int \mathbf{D}\cdot d\mathbf{S}$ (C)

capacitor: $I_d = C\frac{dV}{dt}$ = wire current

plates radius R: $H_\phi = \frac{r}{2}\partial_t D$ $(r < R)$, $\frac{R^2}{2r}\partial_t D$ $(r > R)$; $+\hat{\phi}$ if D_z grows

$\frac{\sigma}{\omega\epsilon} \gg 1$: conductor | $\ll 1$: dielectric

Plane waves

$\partial_z^2 E = \mu\epsilon\,\partial_t^2 E$ | $f(t - \frac{z}{v}) \to +z$ | $g(t + \frac{z}{v}) \to -z$

$v_p = \frac{c}{\sqrt{\mu_r\epsilon_r}}$ | $\eta = \eta_0\sqrt{\mu_r/\epsilon_r}$ | $|\mathbf{E}| = \eta|\mathbf{H}|$

$\mathbf{E} \perp \mathbf{H} \perp$ travel; $\mathbf{E}\times\mathbf{H}$ along travel ($\hat{x}\times\hat{y} = \hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$, $\hat{z}\times\hat{x} = \hat{y}$)

valid wave: depends on $t \mp z/v$ only (or sums); $\cos\omega t\cos\beta z$ = standing

Sheet at $z = 0$

$\mathbf{E} = -\frac{\eta}{2}\mathbf{J}_s\left(t - \frac{|z|}{v}\right)$ both sides: **E even in z**

$\mathbf{E}\times\mathbf{H}$ away from sheet: **H odd in z**, jumps by J_s

course: $\mathbf{J}_s = -J_s(t)\hat{x} \to E_x = \frac{\eta}{2}J_s(t \mp \frac{z}{v})$, $H_y = \pm\frac{1}{2}J_s(t \mp \frac{z}{v})$, $z \gtrless 0$

sheet power $\frac{\eta}{2}J_s^2$/m², half each way ($S = \frac{\eta}{4}J_s^2$)

Reading a wave

vs t at z: same shape, delayed $|z|/v$

vs z at t: shape flipped; front at $|z| = v(t - t_{on})$; **first in time = farthest in z**

snapshot at $t_1 \to t_2$: slide $v(t_2 - t_1)$ along travel

H sketch: E/η, sign from $\mathbf{E}\times\mathbf{H}$; label component + unit

Sinusoids

$\cos(\omega t - \beta z + \phi)$: $\omega = 2\pi f$, $\beta = \frac{2\pi}{\lambda}$, $v = \frac{\omega}{\beta} = \lambda f$

$\Delta\phi$ in $\Delta t \to \omega$; $\Delta\phi$ over $\Delta z \to \beta$

$\mathbf{J}_s = -\cos(\omega t + \phi)\hat{x} \to E_x = \frac{\eta}{2}\cos(\omega t \mp \beta z + \phi)$

Power

$\mathbf{S} = \mathbf{E}\times\mathbf{H}$ (W/m²) | plane wave $S = \frac{E^2}{\eta} = \eta H^2$

$\langle S\rangle = \frac{E_0^2}{2\eta}$ | $w_e = \frac{1}{2}\epsilon E^2 = w_m = \frac{1}{2}\mu H^2$ | $S = v(w_e + w_m)$

$-\oint \mathbf{S}\cdot d\mathbf{A} = \frac{d}{dt}\int(w_e + w_m) + \int \mathbf{J}\cdot\mathbf{E}$

Vector calculus

$\nabla\times\mathbf{F} = \hat{x}(\partial_y F_z - \partial_z F_y) + \hat{y}(\partial_z F_x - \partial_x F_z) + \hat{z}(\partial_x F_y - \partial_y F_x)$

cyl. $\nabla\cdot\mathbf{F} = \frac{1}{r}\partial_r(rF_r) + \frac{1}{r}\partial_\phi F_\phi + \partial_z F_z$

cyl. $\nabla\times\mathbf{F} = \hat{r}\left(\frac{1}{r}\partial_\phi F_z - \partial_z F_\phi\right) + \hat{\phi}(\partial_z F_r - \partial_r F_z) + \frac{\hat{z}}{r}(\partial_r(rF_\phi) - \partial_\phi F_r)$

$\hat{r} = \cos\phi\,\hat{x} + \sin\phi\,\hat{y}$ | $\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$ | $d\mathbf{l} = dr\,\hat{r} + r\,d\phi\,\hat{\phi} + dz\,\hat{z}$

cm² → m²: $\times 10^{-4}$ | $2\pi r$ not πr^2 | ϵ vs ϵ_0 | units: A/m, T, Wb, V, H, F